

Complete Project Overview

Executive Summary

Project Title: Deep Clean?

Genre: First-Person Psychological Horror/Mystery

Platform: PC (Windows)

Engine: Unity 6 (Version 6000.0.28f1) with Universal Render Pipeline

Development Time: 14 weeks

Logline:

You are a cleaner hired to clean out a hoarder's house. It starts off like a normal job but as you dig through the mess, horrific secrets surface. Keep up appearances, collect evidence, and uncover the truth without tipping off the killer.

Synopsis:

You play as a freelance home cleaner that's been hired for a week-long job at an isolated suburban house. At first, the job is like any other yucky hoarder cleanup. But the more you clean, the more disturbing things you discover. A bloody rag. A suitcase of children's toys. A secret room. The deeper you go, the more you suspect the creepy homeowner is hiding a gruesome truth.

You must clean effectively to keep the homeowner satisfied while also secretly collecting and analyzing evidence back at your own home. Missteps raise suspicion. If the owner senses that you're onto him, he may fire you, or worse. Survive the week, solve the mystery, and bring a killer to justice.

Core Concept

Deep Clean is a first-person psychological horror game that subverts the mundane act of cleaning into a tension-filled investigation. Players take on the role of a professional cleaner hired for what appears to be a routine hoarder house cleanup, only to gradually uncover evidence of horrific crimes. The game balances the need to maintain professional appearances with the moral imperative to gather evidence and bring a killer to justice.

Unique Selling Point

- **Mundane Horror:** Transforms everyday cleaning into a suspenseful investigation
- **Moral Tension:** Players must balance job performance with evidence gathering
- **Dynamic Suspicion:** Homeowner AI responds to player behavior, creating emergent tension
- **Detective Mechanics:** Crime board system allows player to piece together clues
- **Multiple Outcomes:** Player choices directly impact the resolution of the mystery

Core Gameplay Functions

1. Cleaning Mechanics

- First person perspective
- Interact with rubbish, grime, furniture, and items to clean and organize
- Tools: mop, spray bottle, rubbish, flashlight
- Clean areas to a certain standard each day to avoid being fired

2. Evidence Discovery System

- As you clean you can discover “suspicious objects” (e.g. photos, stained clothes, locks of hair)
- Use a camera or notepad to discreetly document findings
- Decide whether to leave an object untouched, clean it as normal, or investigate further (risk of raising suspicion)

3. Suspicion System

- A hidden “suspicion meter” track how much the owner distrusts you
- Triggers: lingering near forbidden areas, improperly placing objects, trying to open locked doors, being caught snooping
- If suspicion crosses a threshold, the homeowner reacts differently (e.g. more questions, locks more rooms, watches you)

4. Case Building (Home Hub)

- At the end of each day, return to your apartment
- Review evidence you found and make a “crime board” with pins and string to build theories
- Add your own interpretations. Each theory strengthens or weakens the final case

5. Day/Night Cycle

- Each day starts with a visit to the house (9 – 5)
- After leaving, player goes home to review findings
- MAYBE: late game that includes sneaking back to the house

6. Chapter Structure (7 days)

- Day 1: welcome to the job
 - o Learn controls, basic cleaning
 - o Mildly odd items (expired food, weird smells)
- Day 2: unusual clutter

- First odd discoveries (locked cabinet, odd stains)
 - Meet the homeowner (Friendly but off-putting)
- Day 3: somethings off
 - Personal items that don't belong (kids bag, bloody gloves)
 - Suspicion meter introduces
- Day 4: crossing the line
 - Optional area unlocked. Find disturbing diary entries
 - Homeowner starts watching
- Day 5: uncovering truth
 - Major evidence found (body part, murder tool)
 - Flashbacks at night hinting past trauma
- Day 6: breaking point
 - Suspicion close to max. homeowner aggressive
 - Chance to sneak into basement at night
- Day 7: must finish cleaning, retrieve last clues, avoid detection
 - Final case board review. Decide: confront, call police, or flee

Endings

- Good ending: All evidence found. You report the killer. He's arrested
- Neutral ending: Clean job done, but case incomplete. Killer walks free
- Bad ending 1: Fired. No evidence secured
- Bad ending 2: Suspicion too high. Killed

UI/UX Elements:

- First person HUD: cleaning progress meter, suspicion bar (hidden, only revealed with upgrades or clues)
- Camera and notebook tools
- Inventory for tools and evidence
- Crime board interface with draggable pins and strings

Tone/Visuals:

- Gritty realism. Messy, claustrophobic interiors
- Slow burn psychological horror
- Sound design: subtle audio cues for suspicion, tension, creaks, breathing

Info things:

- Unity 6
 - Universal Render Pipeline (URP)
- Version: 6000.0.28f1
- First Person

Inspo:

- House flipper (mechanics)
- Gone home (narrative discovery)
- Visage/layers of fear (atmosphere)
- Papers, Please (balancing survival and moral tension)

Expansion?

- Additional houses, each with unique crimes.

Game Mechanics

Movement

- WASD – move
- Shift – sprint (limited use)
- Ctrl – Crouch (snooping under/behind things)
- Space – Jump (probably wont need)

Camera/Interaction

- Mouse move – look around
- Left Click – interact (pickup, clean, examine, use tool)
- Right Click – Alt action (e.g. rotate object/zoom/prepare tool)
- Scroll Wheel – change cleaning tool or zoom in/out
- E – examine closer (hold or tap)
- F – flashlight toggle (maybe?)

Inventory/Tools

- 1-4 – switch tools
- Q – open radial tool menu (maybe?)
- Tab – open evidence notebook
- I – inventory (manage objects and clues)

Crime Board (at home only)

- Tab or M – Open case board
- Drag & drop with mouse – Arrange clues
- Right Click – Add connection or remove
- ESC – Exit board / return to apartment

Special Features

- C – take a photo

- R – review photos (At home only)
- T – Take notes (simple typing interface)

Homeowner Encounters

- ESC – open dialogue options / back away (non-confrontational)
- Left Click – Respond/interact (if I decide to do conversations)

Narrative Structure and Themes

Core Themes

- **Trust and Deception:** Surface appearances vs. hidden truths
- **Moral Responsibility:** When to act on suspicion vs. professional duty
- **Class Dynamics:** Working-class protagonist investigating wealthy homeowner
- **Justice vs. Safety:** Personal risk in pursuing truth

Story Arc Overview

The narrative follows a classic mystery structure with horror elements:

1. Establishment: Normal cleaning job setup
2. Inciting Incident: First suspicion discovery
3. Rising Action: Escalating evidence and suspicion
4. Climax: Major revelation and confrontation risk
5. Resolution: Multiple endings based on player choices and evidence collected

Art Direction and Visual Style

Art Style Overview

Low-Poly Stylized Realism: Combines the accessibility of low-polygon modeling with realistic lighting and materials to create an unsettling, uncanny valley effect that enhances the psychological horror elements

- Low poly meshes: 500 – 1500 tris ?
- Decals: for blood, grime, dirt, instead of baking it into textures. Decal Master (assets store), Decal System in URP

Color Palette and Mood

- **Primary Palette:** Desaturated earth tones (browns, grays, muted yellows)
- **Accent Colors:** Sickly greens and rust reds for suspicious elements
- **Lighting Temperature:** Cool daylight gradually warming to create unease

- **Post-Processing:** Heavy vignetting, subtle film grain, chromatic aberration for tension



Atmosphere – Paratopic (ps1), Gone Home, What remains of Edith Finch



Shader Setup:

1. Lit Shader (base shader) – unitys URP/Lit shader with changes:
 - a. Roughness/smoothness tuned low so not glossy
 - b. Desaturated, grimy textures

- c. Use vertex colors for grime masks (maybe?)
- 2. Grime/blood overlay shader
 - Tiling detail texture (e.g. dirt map)
 - Optional mask support (for decals or grime areas)
 - Blend mode: multiply or overlay

Tools:

- Shader Graph in URP
 - o Base Lit Shader + Detail Overlay (Grime map + mask)
 - o Add params to control dirtiness over time (for interactive grime reveals)
- 3. Post processing setup

With unity URP's Volume system:

- Color grading: desaturate, lower contrast, add subtle green/blue tints
- Vignette: strong vignette adds claustrophobic feel
- Film grain: subtle but creepy
- Depth of field: slight blur help sell tension and focus
- Bloom: only for lights (adds surreal feel)
- Chromatic aberration: only at high suspicion or in key moments

Helpful Tools:

- ProBuilder – for level shaping
- Snap Art Assets (in asset store) – Modular furniture, walls, props
- Post Processing (URP built in) – just add a global volume

Audio Design Philosophy

Atmospheric Audio

- **Environmental:** Creaking floors, settling house sounds, distant traffic
- **Interactive:** Satisfying cleaning sound effects, object placement audio
- **Tension:** Subtle heartbeat during high suspicion, breathing changes
- **Dynamic:** Audio cues that respond to suspicion level and discoveries

Implementation Strategy

- **3D Spatial Audio:** Full surround sound support for immersion
- **Procedural Elements:** Dynamic audio mixing based on player state
- **Music:** Minimal ambient drones, tension strings for key moments

- **Voice:** Text to speech placeholder with upgrade path to voice acting

Capstone Project Scope

Included Features

- One day/night cycle with opportunity for more day progression
- Core cleaning mechanics with professional tools
- Evidence discovery and documentation system
- Dynamic suspicion AI system
- Crime board investigation interface
- Multiple endings based on player performance
- Atmospheric audio design and post-processing effects

Primary Role Responsibilities

Game Designing and System Programming (40%)

Core System Design

Suspicion Mechanics System:

- Design mathematical formula for suspicion accumulation/decay
- Create trigger conditions and thresholds for homeowner reactions
- Balance risk/reward ratios for evidence gathering vs job performance
- Implement state machine for homeowner AI behavior changes

Evidence Discovery System:

- Design evidence placement logic and interaction hierarchies
- Create photographing and note-taking mechanics
- Develop evidence categorization and storage system
- Balance evidence accessibility vs. player exploration skill

Cleaning Mechanics Implementation:

- Program tool switching and usage systems
- Create satisfaction meters and completion tracking
- Design interactive object physics and state changes
- Implement area completion algorithms and visual feedback

Technical Implementation Tasks:

- Player controller programming (movement, interaction, tool usage)
- Save/load system for evidence, progress, and game state
- Day/night cycle transition programming
- UI state management and data persistence

Level Designing and Environmental Storytelling (25%)

House Layout and Flow Design

Spatial Design Responsibilities:

- Create detailed floor plans for 1-story house layout
- Design room progression and unlocking sequences
- Plan evidence placement for optimal discovery pacing
- Implement environmental story telling through object placement

Level Implementation Tasks:

- Greybox house layout in unity
- Lighting setup and atmosphere creation for each room
- Interactive object placement and trigger zone configuration
- Secret area and hidden space implementation

Pacing and Player Guidance

- Design visual and audio cues to guide player attention
- Create natural exploration paths that feel organic
- Balance obvious vs. hidden evidence placement
- Plan camera angle opportunities for dramatic reveals?

Narrative Design and Mystery Construction (20%)

Story Structure Development

Mystery Construction:

- Design overarching murder mystery with logical evidence chain
- Create homeowner character background and motivation
- Develop multiple victim profiles and timeline of events
- Write evidence descriptions and diary entries for environmental storytelling

Multiple Ending Design:

- Script Conditional ending sequences based on evidence collected
- Design confrontation dialogue trees and player choice consequences
- Create police report outcomes based on case strength
- Write flavor text for crime board theories and connections

Character Development

- Homeowner personality traits and behavioral patterns

- Create player character void through internal monologue and notes
- Develop victim characterization through discovered belongings
- Write atmospheric text for item examinations and discoveries

Technical Art and Shader Developing (10%)

Visual System Implementation

Custom Shader Development:

- Create grime overlay shader system for environmental storytelling
- Implement decal system for blood, dirt, and wear patterns
- Design material blending system for cleaning state transitions
- Develop post-processing effect chains for tension and atmosphere

Audio Design and Implementation (5%)

Atmospheric Audio Creation

Sound Design Responsibilities:

- Record and edit cleaning tool sound effects (vacuum, mop, spray)
- Create environmental ambience for house atmosphere
- Design tension audio cues for suspicion level changes
- Implement 3D spatial audio for immersive environmental storytelling

Audio programming:

- Script audio trigger systems for discoveries and interactions
- Create adaptive music system that responds to player state
- Implement audio mixing profiles for different tension levels
- Program audio feedback systems for player actions

Risk Assessment and Mitigation

High-Risk Elements

1. **Balancing Suspicion Mechanics:** Risk of frustrating players
 - a. Mitigation: Extensive playtesting and adjustable difficulty
2. **Maintaining Horror Atmosphere:** Without relying on jump scares
 - a. Mitigation: Focus on psychological tension and sound design
3. **Evidence System Complexity:** Players missing crucial clues
 - a. Mitigation: Multiple evidence paths and optional hint system

Medium-Risk Elements

1. **Scope Creep:** Adding features beyond core mechanics
2. **Art Asset Creation:** Consistent quality across all models and textures

Low-Risk Elements

1. **Basic Gameplay Mechanics:** Well established interaction patterns
2. **Audio Implementation:** Standard 3D audio requirements

Overall Project Goals

Primary Objectives

- **Complete Playable Experience:** Deliver a fully functional psychological horror cleaning game with at least one complete day/night cycle
- **Core Mechanics Implementation:** Successfully implement cleaning, evidence discovery, and suspicion systems
- **Atmospheric Excellence:** Create compelling psychological tension through environmental storytelling and audio design.
- **Multiple Endings:** provide meaningful player choice consequences with at least 3 different endings.
- **Technical Proficiency:** Demonstrate advanced Unity programming and unique game system designs.

Success Metrics

- **Gameplay Flow:** Players can complete one full day without confusion about objectives.
- **Tension Building:** Measurable increase in player engagement during evidence discovery moments.
- **Mystery Coherence:** All evidence pieces logically connect to solve the central mystery
- **User Experience:** Intuitive controls and clear visual feedback for all interactions

Critical Success Factors:

- Maintain focus on psychological horror experience
- Ensure evidence discovery remains central to gameplay
- Keep technical complexity manageable within timeline
- Prioritize player clarity and intuitive interactions
- Balance atmospheric tension with gameplay accessibility

Capstone Deliverables and Timelines

Week 1-4: Foundation Systems

Personal Deliverables/Milestones:

- Complete player controller with tool switching
- Basic cleaning mechanics (mop, vaccum, spray bottle)
- House layout with room progression
- Core suspicion system mathematical model

Skills used:

- C# programming proficiency
- Unity physics system understanding
- Game Balancing and mathematical modeling
- 3D spatial design capabilities

Week 5-8: Content Creation Sprint**Personal Deliverables/Milestones:**

- Complete evidence discovery system with photography
- Decided mystery content (evidence, diary entries, environmental details)
- Crime board interface with drage and drop funcitonality
- Basic homeowner AI with mutiples behavior states

Skills demonstrated:

- UI/UX design and implementation
- Creative writing and narrative design
- AI programming and state machine design
- Content creation and asset management

Week 9-12: Polish and Integration**Personal Deliverables/Milestones:**

- Complete art asset integration and material setup
- Sahder implementation for atmosphere and grime
- Audio implementation with 3D spatial positioning
- Save/load system with evidence persistence

Skills demonstrated:

- Technical art and shader programming
- Audio implementation and mixing
- Data management and serialization
- Performance optimization and profiling

Week 13+: Testing and Finalization**Personal Deliverables:**

- Different possible endings implementation
- Complete testing suite and bug resolution
- Performance optimization for target hardware specifications
- Final polish pass with particle effects and post-processing

Skills demonstrated:

- Quality assurance and systematic testing
- Performance profiling and optimization
- Project management and milestone delivery
- User experience testing and iteration

Learning Objectives and Skill Development

Technical Skills to Demonstrate

1. **Advanced Unity Programming:** State machines, custom editors, performance optimization
2. **Gameplay Systems Design:** Balancing mechanics, player psychology, risk/reward systems
3. **Environment Storytelling:** Visual narrative techniques, atmospheric design
4. **Audio Implementation:** 3D spatial audio, dynamic mixing, procedural audio cues
5. **Shader Programming:** Custom materials, post-processing effects, visual atmosphere

Creative Skills to Showcase

1. **Mystery Writing:** Logical plot construction, evidence placement, character development
2. **Level Design:** Player flow, exploration pacing, environmental storytelling
3. **Game Balancing:** Playtesting interpretation, mathematical modeling, difficulty curves
4. **Visual Design:** Art direction consistency, mood creation, technical art pipeline
5. **User Experience Design:** Interface design, player guidance, accessibility considerations